SILICYTE

Does AI self-improvement actually compound?

Pre-registered, compute-matched tests of recursive self-improvement, with the test set sealed and every result published, including nulls.

Braxton Mensah, founder · silicyte.com · hello@silicyte.com · November 2026

THE PROBLEM

Self-improvement claims are graded on the tests they were selected on.

- AI systems now write the tools, prompts and code their next round runs on.
- Published library-learning gains largely disappeared once compute was matched and reuse was measured (Berlot-Attwell et al., arXiv 2410.20274 and 2504.03048).

THE STUDY: SILICYTE RATCHET (SILICYTE-0119, REGISTERED 4 NOV 2026)

Does a growing library of verified helpers beat its look-alikes at equal tokens?

Arm	What it rules out
Frozen after round 1	gain without growth
Same tokens, one round	dose, not compounding
Verified examples, no helpers	context, not reuse
Matched sham library	prompt length and genre
Extra samples, no library	plain test-time compute

400 sealed ARC-AGI-1 tasks (id list hashed before any run), a reuse gate, a knockout, a contamination gate, and a planted positive control that voids the run if the harness cannot

see a real library. Pre-registration: github.com/braxtonmensah/ratchet. Result pending; our registered prediction is **+2** of 400.

WHY TRUST US

We retract our own best numbers in public.

- Our earlier headline, a guardrail hook absent on **29.8%** of the calls it existed to check, came out a median of **0.0000** on clean public machines. We said so on the homepage.
- Paper: degrading a design loop's scorer made its reported gain 2.4x larger (t = **-14.60**; DOI 10.5281/zenodo.23441786).

FOUNDER AND STAGE

Solo, technical, pre-revenue.

- Braxton Mensah, Indiana University undergraduate. Directs an AI-agent research fleet. Trained and served his own ARC Prize 2026 system; official results 4 Dec 2026.
- $0 revenue, 0 customers, unincorporated, no money raised.

Design and implementation are AI-assisted; the founder sets the questions, the registered pass rules and the retractions. Not peer reviewed.